Clinical-Genomics Harmonisation and AI Baseline Module

TCGA-BRCA | Updated Documentation

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1. Background

This module was undertaken as part of a Month 1 internship task at Multiomics Intelligence (MOI), with the goal of building a reproducible pipeline that harmonises clinical and somatic mutation data from TCGA (The Cancer Genome Atlas) for downstream machine-learning and survival-analysis applications.

A review of the initial version of this work identified issues that required correction before the pipeline could be considered analysis-ready: the five-year survival target variable had been computed on the wrong denominator (censored patients had been counted as "survived"), only three ML models had been trained, a data-leakage issue existed in the feature-scaling step (scaling was fitted before the train/test split) and the language used to describe genomic patterns needed to be made more cautious. This document supersedes the earlier version and reflects the corrected pipeline (Scripts 01–09).

2. Aim

- Build a harmonised clinical–genomic dataset from TCGA-BRCA suitable for AI/ML baseline modelling.
- Correctly define a five-year survival outcome using only patients with confirmed status, rather than including censored patients as "survived."
- Train and compare five baseline ML classifiers (Logistic Regression, Elastic Net, Random Forest, XGBoost, SVM) without data leakage, evaluated on a held-out test set.
- Evaluate whether the trained model generalises to an independent external dataset (CPTAC-BRCA 2020).

3. Dataset Selection

Three TCGA cohorts were reviewed for suitability (Script 01): TCGA-BRCA, TCGA-LUAD, and TCGA-LAML.

Metric	TCGA-BRCA	TCGA-LUAD	TCGA-LAML
Clinical patients	1,098	585	200
Unique patients (MAF)	965 (964 usable)	557	199
AGE available	Yes (16 missing)	Yes (94 missing)	NO — all missing
AJCC Stage available	Yes (12 missing)	Yes (71 missing)	NO — all missing
OS_STATUS	945 alive / 152 deceased	334 alive / 188 dead / 63 NA	67 alive / 133 deceased
Top gene (raw count)	PIK3CA (367 muts)	TTN — likely passenger	DNMT3A (77 muts)

TCGA-BRCA was selected as the primary dataset for this module: it is the largest cohort, has the most complete clinical annotation (age, stage, overall survival, disease-free survival), and uses standard AJCC staging. TCGA-LUAD was set aside

for this task (smaller cohort; 10.8% of patients missing OS_STATUS; the top raw-count gene, TTN, is a known passenger). TCGA-LAML remains a candidate for cross-cancer generalisability testing in a later phase, though age and staging are entirely unavailable for that cohort.

4. Primary Dataset Overview — TCGA-BRCA

Data source: cBioPortal (TCGA-BRCA, GDC-harmonised calls). Pipeline language: R (tidyverse, maftools, survival, survminer, glmnet, randomForest, xgboost, e1071).

Stage	N patients	% of total	N deceased	% deceased	Mean OS (months)
Stage I	183	16.7%	16	8.7%	48.8
Stage II	621	56.6%	66	10.6%	39.5
Stage III	249	22.7%	44	17.7%	35.3
Stage IV	20	1.8%	15	75.0%	34.3
Other/NA	25	2.3%	—	—	—

Stage IV patients show markedly higher mortality (75.0%) than earlier stages, consistent with expected clinical progression. Deceased patients show a higher mean OS than patients still alive at last contact; this reflects right-censoring — OS_MONTHS for living patients records time-to-last-contact, not total lifespan, so it should not be read as a survival advantage.

5. Methods

5.1 Clinical data cleaning (Script 02)

The clinical patient (1,098 × 31) and sample (1,102 × 10) files were standardised into 21 analysis-ready columns, including os_event, os_months, stage_clean, stage_numeric, path T/N/M stage, year of diagnosis, and primary diagnosis.

- Stage strings (e.g. "Stage IIIA") were parsed by checking IV → III → II → I, in that order of specificity. Checking "Stage II" before "Stage III" would misclassify Stage III patients, since "Stage IIIA" contains the substring "Stage II" — a known TCGA cleaning pitfall.
- os_event was derived as 0 = alive, 1 = deceased from OS_STATUS; survival_days was taken from DAYS_TO_DEATH where deceased, otherwise DAYS_LAST_FOLLOWUP.

5.2 Genomic data processing (Script 03)

The raw MAF file (89,012 mutation records, 140 columns) was filtered to coding, protein-altering variant types (missense, nonsense, frameshift, splice-site, in-frame indels, nonstop, translation-start-site), and Silent/intronic/UTR/flanking/IGR calls were removed as non-coding.

Metric	Value
Total raw mutations	89,012
Coding mutations (kept)	66,687 (74.9%)
Non-coding removed	22,325 (25.1%)
Unique patients with mutations	964
Mean / median TMB (mut/Mb)	1.82 / 0.87
Hypermutated (TMB > 10)	17 patients (1.8%)

5.3 Patient/sample ID harmonisation (Script 04)

Patient IDs were extracted from the first 12 characters of the MAF tumour-sample barcode and joined against the clinical table. Of 1,098 clinical patients, 964 (87.8%) had at least one coding mutation and entered the merged dataset; 134 patients had clinical data but no genomic data (not all TCGA patients were sequenced — common reasons include insufficient tumour material or low tumour cellularity). All genomically profiled patients had matching clinical records. The merged clinical-genomic dataset contains 964 patients and 34 columns.

5.4 Summary statistics and visualisation (Scripts 05–06)

Descriptive summaries and mutation-landscape figures were generated for the 964-patient merged cohort, including top mutated genes overall and by stage, variant classification breakdown, tumour mutation burden by stage, and standard maftools diagnostics (oncoplot, Ti/Tv ratio).

Two co-dominant driver genes, TP53 and PIK3CA, were each mutated in roughly a third of patients; a pattern

consistent with the two genes marking largely distinct molecular subtypes has been reported previously — TP53 alterations are more frequent in triple-negative/basal-like tumours, PIK3CA alterations in luminal subtypes — though subtype assignment was not performed directly in this module.

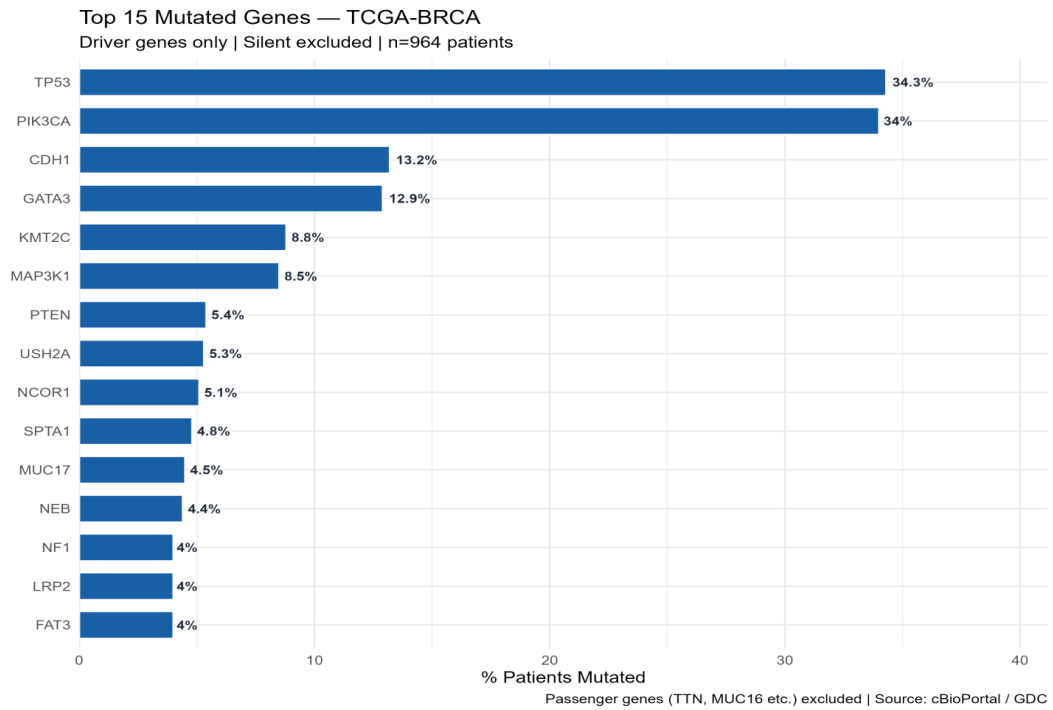


Fig. 5.4a — Top 15 coding-mutated driver genes, TCGA-BRCA (n=964). TP53 (34.3%) and PIK3CA (34.0%) are co-dominant.

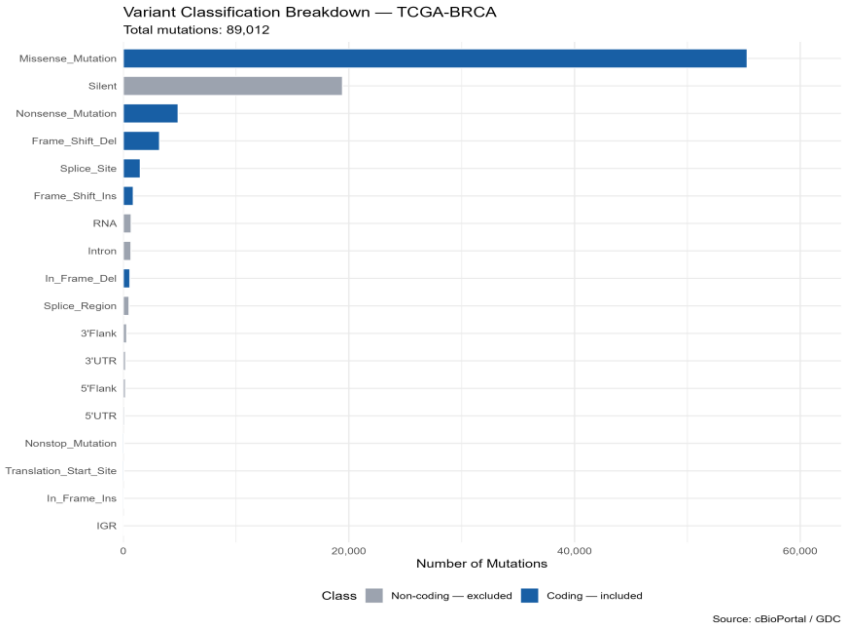


Fig. 5.4b — Variant classification breakdown. Blue = coding (retained); grey = non-coding (excluded).

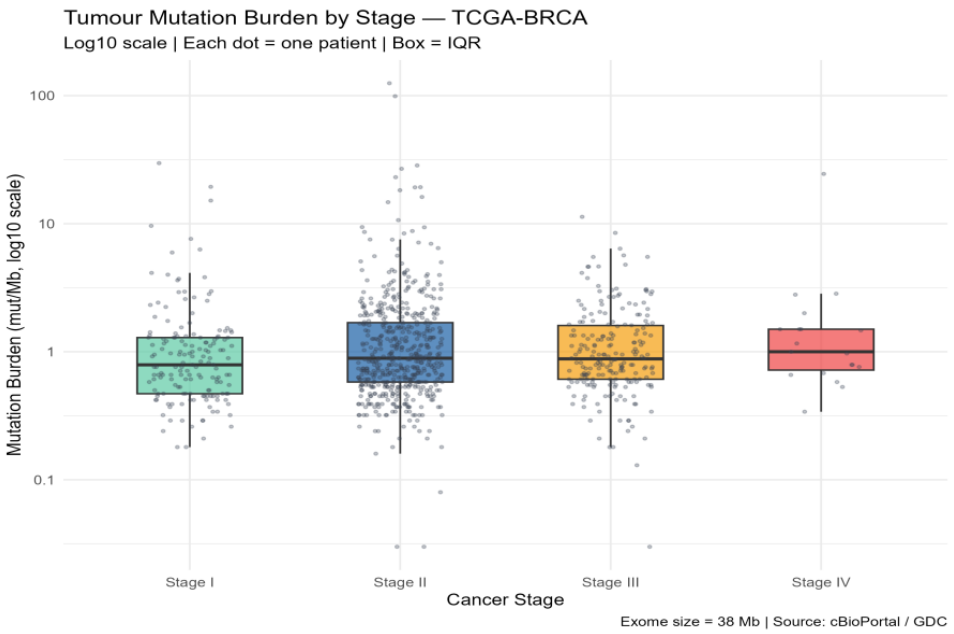


Fig. 5.4c — Tumour mutation burden (log10 mut/Mb) by AJCC stage. No clear monotonic trend with stage, consistent with published BRCA TMB literature.

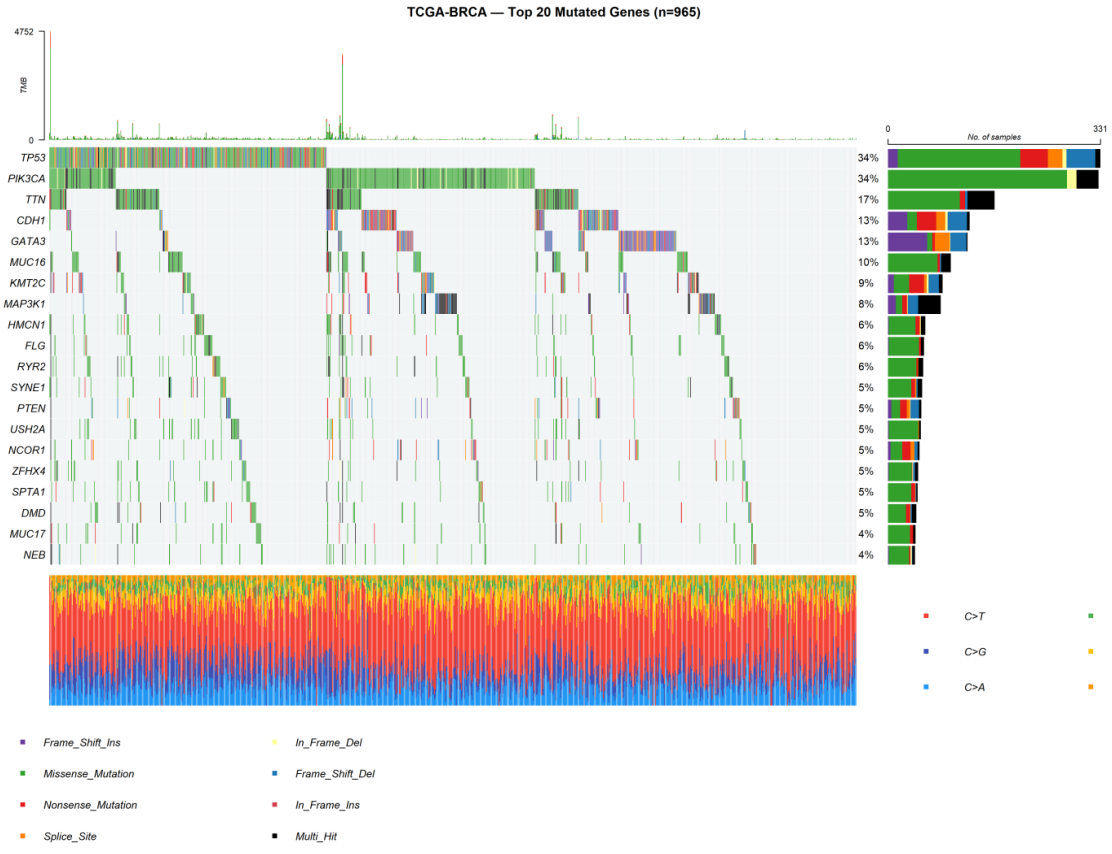


Fig. 5.4d — Oncoplot of the top 20 mutated genes across all profiled patients (n=964).

5.5 ML-ready feature table construction (Script 07)

The five_year_survival target was engineered from os_event and os_months, since no direct survival-status column exists in the raw clinical data:

Case	Condition	five_year_survival	Meaning
1	os_event = 0 (alive at last contact)	= 1	Right-censored, alive — treated as survived
2	os_event = 1 AND os_months ≥ 60	= 1	Died after 5 years — confirmed survived to 5y
3	os_event = 1 AND os_months < 60	= 0	Confirmed did not survive 5 years
Excluded from ML	os_event = 0 AND os_months < 60 (censored before 5y), or missing OS data	= NA	Outcome genuinely unknown at 5 years — cannot be labelled either way

The earlier version of this pipeline incorrectly folded Case-4 (alive/censored with less than 60 months of follow-up) into the "survived" class, since a patient who is alive but has only been followed for, say, 20 months has not actually been confirmed to survive to 5 years. Correcting this reduces the ML-eligible cohort from 964 to 305 patients — those with a genuinely confirmed 5-year outcome. The remaining 659 censored patients are excluded from this module's modelling; their outcomes are not yet resolved and are not usable for the classification task defined here.

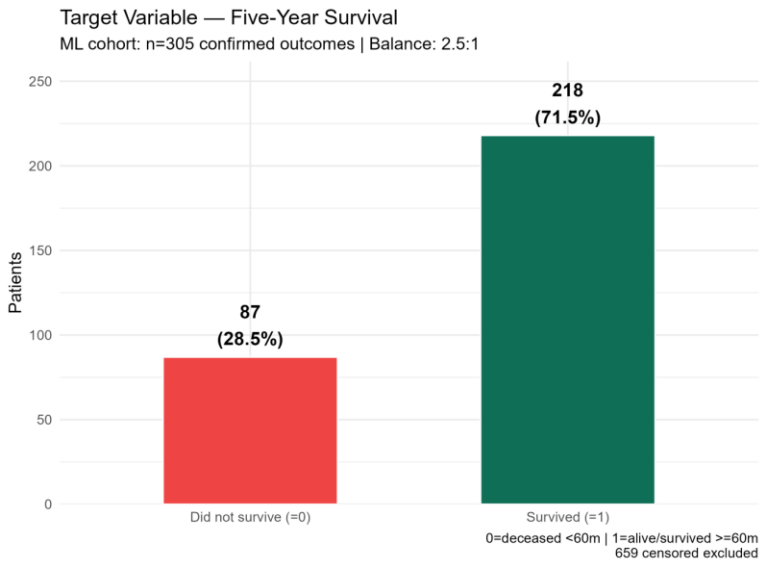


Fig. 5.5a — Corrected target variable distribution. ML cohort: n=305 confirmed outcomes; 218 survived (71.5%), 87 did not (28.5%); class ratio ≈2.5:1.

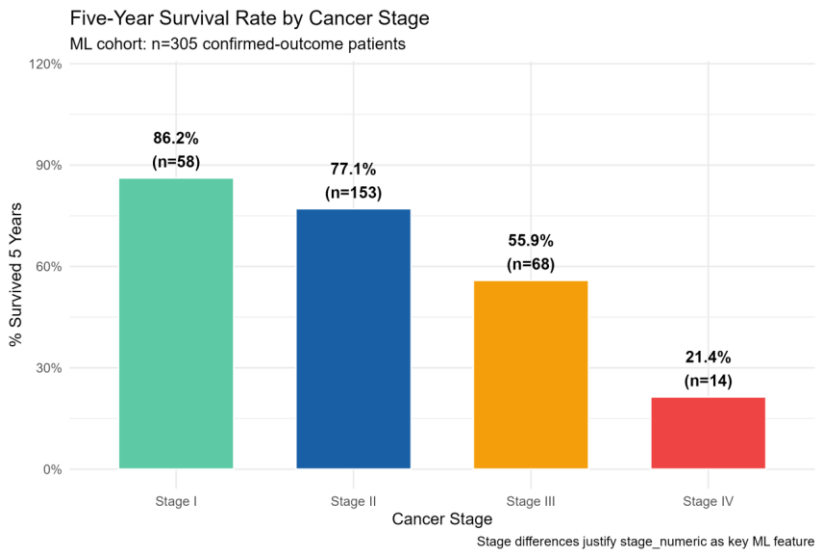


Fig. 5.5b — Five-year survival rate by cancer stage within the confirmed-outcome cohort (n=305): Stage I 86.2% (n=58), Stage II 77.1% (n=153), Stage III 55.9% (n=68), Stage IV 21.4% (n=14).

The final feature table combines 6 clinical features (age, stage_numeric, one-hot-encoded primary diagnosis), 6 genomic summary features (tmb_nonsynonymous, n_truncating, n_nonsense, n_frameshift_del, n_frameshift_ins, n_splice_site), and 20 binary gene-mutation flags (the top 20 most frequently mutated genes, included without pre-filtering for driver/passenger status so that the models themselves can weigh relevance).

5.6 Baseline ML models (Script 08)

Five classification models were trained on the 305-patient confirmed-outcome cohort: Logistic Regression, Elastic Net (glmnet, alpha=0.5), Random Forest, XGBoost, and a kernel SVM (e1071).

Data-leakage prevention: the 305-patient cohort was split 80/20 (stratified, seed = 42) before any scaling was applied. The scaler (mean/SD) was fitted on the training set only and then applied to the test set; binary gene-flag and one-hot columns were left unscaled. This corrects the earlier version of the pipeline, in which all 964 patients were scaled together before the split — a data-leakage error, since information from the test set could influence the scaling parameters used on the training set.

Split	N patients	Survived (1)	Did not survive (0)	Method
Training (80%)	≈244	≈174 (71%)	≈70 (29%)	Stratified random split, seed = 42
Test (20%)	62	44 (71%)	18 (29%)	Held out — not used during training or scaling

5.7 External dataset test — CPTAC (Script 09)

The CPTAC-BRCA 2020 cohort (122 patients with clinical data, 122 with matched mutation data; 19,548 coding mutations) was evaluated as an independent external test. All 20 training genes were present in CPTAC (20/20 overlap).

A systematic variable-availability check (Script 09a) found that CPTAC provides AGE (86.1% complete) but does not provide OS_STATUS, OS_MONTHS, AJCC stage, or primary diagnosis — the columns needed to compute the five_year_survival outcome or to evaluate AUC on this cohort.

Variable needed	Found in CPTAC?	Coverage
OS_STATUS	Not found	0%
OS_MONTHS	Not found	0%
AGE	Found	86.1% (105/122)
PATH_STAGE / AJCC stage	Not found	0%
Primary diagnosis	Not found	0%

Given this, Script 09b ran the pipeline in prediction-only mode: the five trained models (plus an unweighted ensemble average) were applied to CPTAC patients to generate predicted 5-year-survival probabilities, without computing AUC, recall, or specificity. There is no ground-truth outcome available in CPTAC against which to score those metrics. Missing stage was imputed as Stage II (the training-set median stage); missing diagnosis one-hot columns were set to 0 (i.e., no diagnosis category flagged). This is a genuine, disclosed limitation of the external test rather than a result to be read as validation accuracy.

6. Results

6.1 Five ML models — test set (n=62), confirmed-outcome cohort

Model	AUC	Accuracy	Recall	Precision	Specificity	F1	MCC
Elastic Net	0.815	0.790	1.000	0.772	0.278	0.871	0.463
XGBoost	0.792	0.710	1.000	0.710	0.000	0.830	0.000
SVM	0.780	0.790	0.977	0.782	0.333	0.869	0.446
Logistic Regression	0.726	0.661	0.864	0.717	0.167	0.784	0.039
Random Forest	0.725	0.710	0.932	0.732	0.167	0.820	0.151

Elastic Net achieved the best overall discrimination (AUC = 0.815) and, unlike XGBoost, retains some ability to identify non-survivors (specificity = 0.278 vs 0.000). XGBoost achieved a competitive AUC (0.792) but, at the default 0.5 threshold, classified every test patient as a survivor, correctly identifying none of the 18 true non-survivors — a class-imbalance-driven failure mode also seen with Logistic Regression and Random Forest at lower specificity. SVM offered the best specificity (0.333) among all five models while maintaining a strong AUC. This pattern is consistent with a class-imbalanced dataset (71.5% vs 28.5%) in which most classifiers gravitate toward predicting the majority class unless explicitly penalised or thresholds are adjusted.

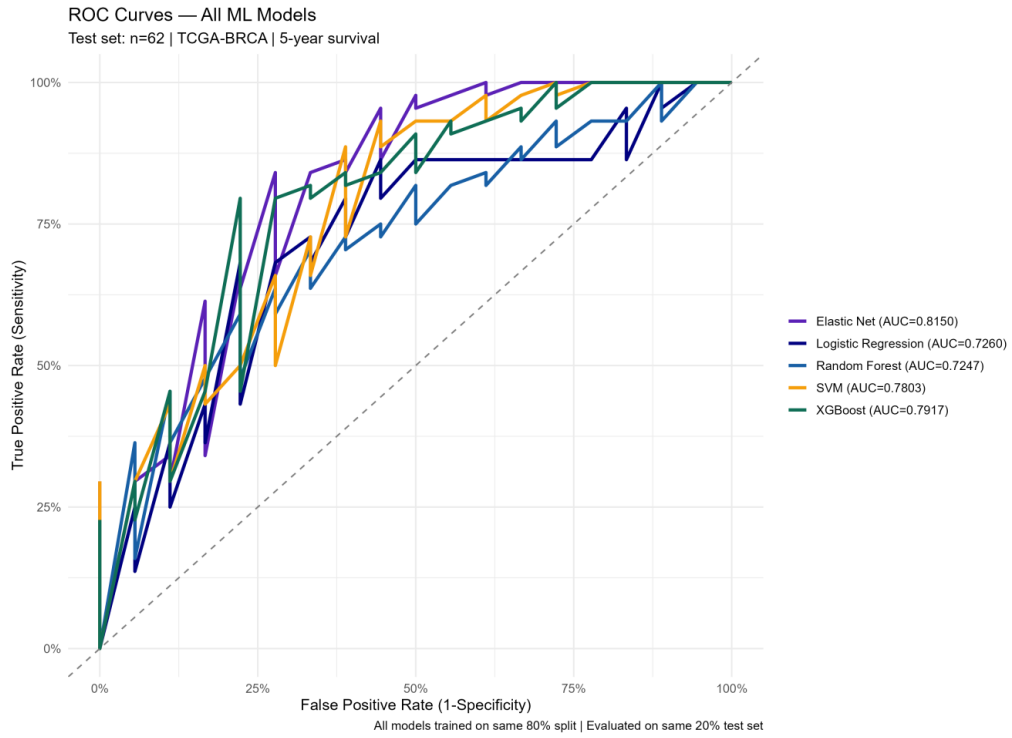


Fig. 6.1a — ROC curves for all five models on the held-out test set (n=62).

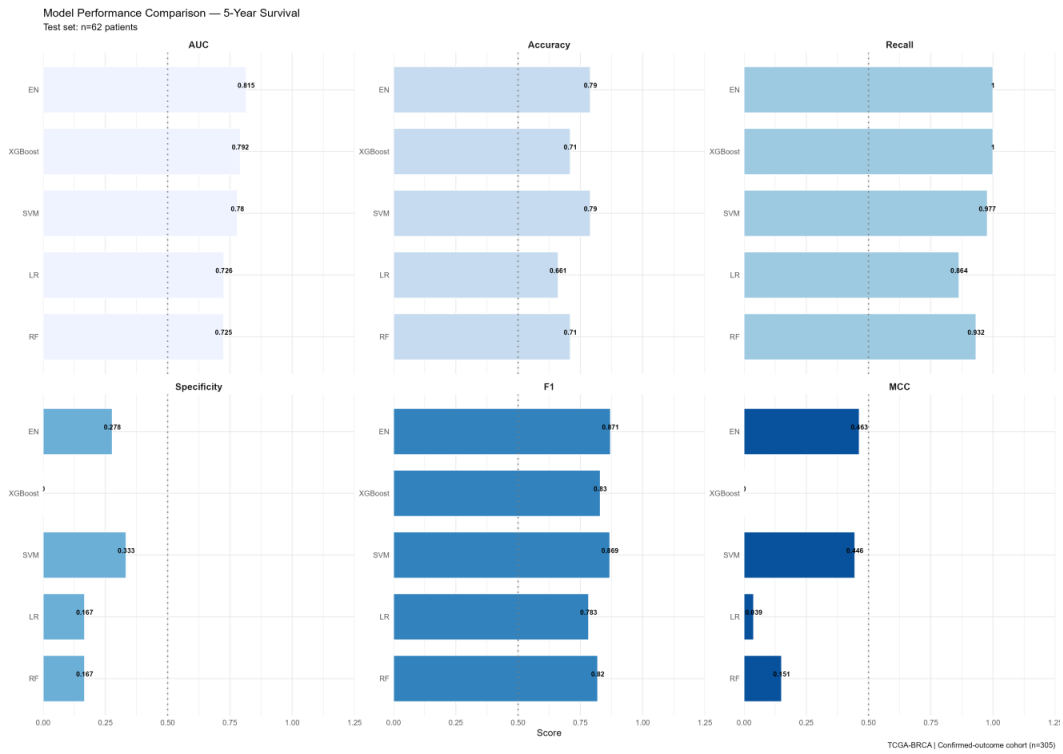


Fig. 6.1b — Model performance comparison across AUC, accuracy, recall, specificity, F1, and MCC.

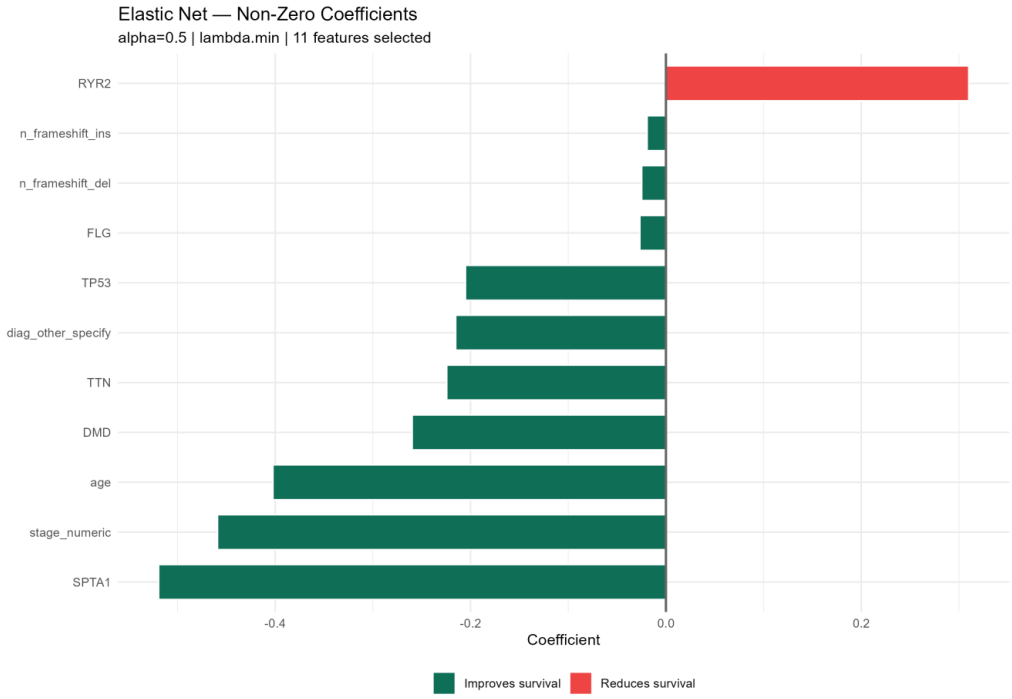


Fig. 6.1c — Elastic Net (best-performing model) non-zero coefficients — 11 features selected. Stage and age are the strongest predictors of reduced 5-year survival; several individual gene flags (e.g. SPTA1, TP53, TTN, DMD) contribute smaller effects in this penalised model.

6.2 External test — CPTAC-BRCA 2020 (prediction only)

The trained models were applied to 122 CPTAC patients to generate predicted probabilities of 5-year survival. As no ground-truth outcome is available in CPTAC, these results are reported as predicted risk distributions rather than validation accuracy.

- 72 of 122 patients (59.0%) fell into the "high survival predicted" category (ensemble probability ≥ 0.75); 50 of 122 (41.0%) fell into "moderate survival predicted" (0.50–0.75). No CPTAC patient was predicted below the 0.5 decision threshold by the ensemble.
- Model agreement was generally high: in 96 of 122 patients (78.7%), all five individual models agreed on the predicted class; in 19 patients, four of five agreed; in 6 patients, three of five agreed; and in 1 patient, only two of five agreed.
- Predicted survival probability declined gradually with age at diagnosis (LOESS trend), consistent with the age effect estimated by the ML models on TCGA-BRCA.

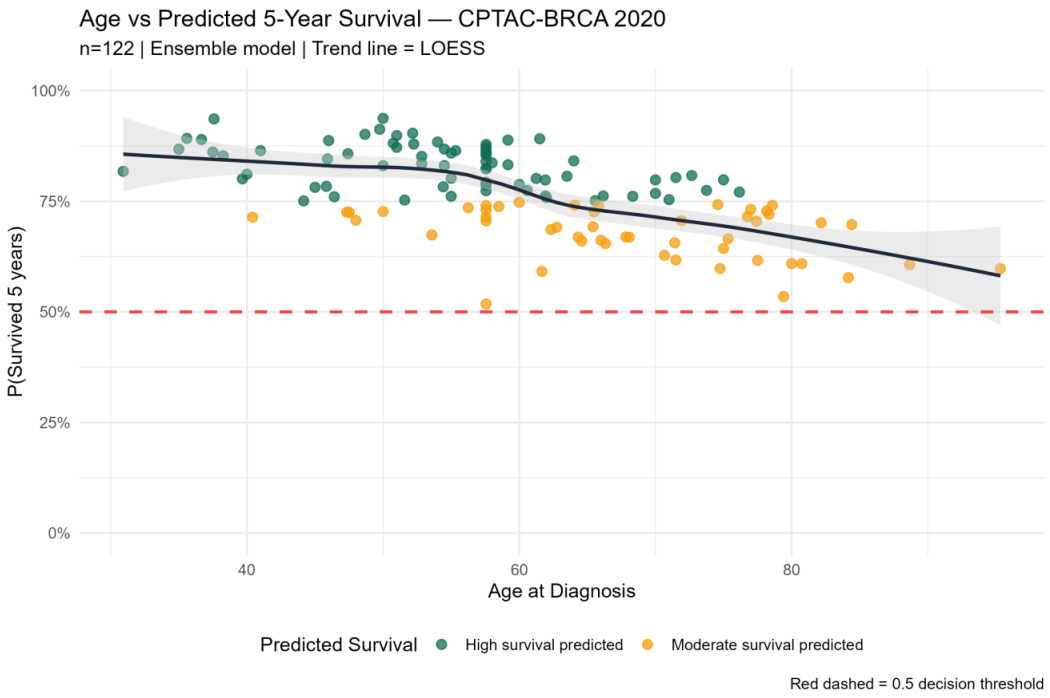


Fig. 6.2a — Predicted 5-year survival probability vs. age at diagnosis, CPTAC-BRCA 2020 (n=122, ensemble model).

These findings are consistent with the model transferring its learned age/stage-driven risk pattern to an independent cohort, but this cannot be described as validated performance: without CPTAC outcome data, it is not possible to compute AUC, sensitivity, or specificity on the external set, and the results should be read as a plausibility check rather than confirmation of predictive accuracy.

7. Key Figures

All figures referenced above are numbered by the section in which they first appear. For completeness, the outputs directory also contains per-model ROC curves, confusion matrices, and feature-importance plots for each of the five ML models individually (outputs/TCGA-BRCA/figures/fig08_[model]_*.png), which are available on request but are not reproduced in full here for brevity.

8. Model Performance Summary

Analysis	Cohort	Key metric	Result
5 ML models (best: Elastic Net)	n=305 (confirmed outcomes), 20% test = 62	AUC	0.815 (Elastic Net)
CPTAC external application	n=122 (prediction only)	N/A — no ground truth	59.0% high-survival predicted; 41.0% moderate

9. Interpretation

- Stage and age are consistently the dominant predictors of 5-year survival across the ML models used in this module both the feature-importance rankings and the Elastic Net coefficients agree on this point.
- Genomic features (TMB, specific gene flags, mutation-type counts) contribute secondary, smaller signal on top of the clinical features, rather than dominating prediction consistent with the modest cohort size (305 confirmed outcomes) relative to the number of candidate gene features.
- Elastic Net's advantage over the other four models is plausibly explained by its combination of regularisation (which controls overfitting given the moderate sample size) and retained ability to separate classes, whereas XGBoost, despite a similar AUC, collapsed to predicting the majority class at the default threshold that is a reminder that AUC alone does not capture threshold-dependent clinical usability.
- The correction to the survival target (964 → 305 confirmed-outcome patients) substantially changes the class balance (from 91%/9% to 71.5%/28.5%), which in turn changes which metrics are informative; specificity and MCC, which were uninformative (0.000) under the old class balance, now differentiate model quality meaningfully.

10. Limitations

- The confirmed-outcome ML cohort (n=305) is modest for training 20+ gene-flag features; individual gene coefficient estimates (e.g. SPTA1, DMD, TTN in the Elastic Net model) should be treated as exploratory.
- The 659 censored patients excluded from the five_year_survival target (patients alive with less than 60 months of follow-up) are not used anywhere in this module's modelling; their outcomes remain genuinely unresolved at the 5-year mark.
- CPTAC-BRCA 2020 lacks survival, staging, and diagnosis data entirely, so external validation in the formal sense (AUC on ground truth) was not possible in this module; the CPTAC results reported here are predicted risk distributions only.
- Stage was imputed as "Stage II" (the TCGA-BRCA training median) for all CPTAC patients, since no staging data exists there; this is a simplification that likely understates real heterogeneity in the external cohort and should be kept in mind when interpreting the CPTAC predictions.

11. Future Prospective

- Investigate class-imbalance mitigation (SMOTE, class weighting, or threshold tuning) to improve specificity for models beyond Elastic Net.
- Extend the pipeline to TCGA-LUAD and/or TCGA-LAML to test cross-cancer generalisability, noting the missing age/stage fields for LAML.
- Identify or request an external cohort with complete survival annotation for a formal (AUC-based) external validation of the survival classifier.